

# Thompson & Cosmelli: Why the Vat Must Be a Surrogate Body

## Overview

Evan Thompson and Diego Cosmelli develop one of the clearest enactive critiques of the brain-in-a-vat scenario. Across a 2010 chapter and a closely related 2011 journal article, they argue that a genuinely functional vat would have to do far more than passively keep a brain alive. It would need to regulate the brain's metabolism, preserve the body's homeodynamic loops, and support active sensorimotor coupling with an environment. Once those requirements are taken seriously, the vat ceases to be a mere container and becomes a surrogate body.[cite:4][cite:7][cite:10]

Their key conclusion is explicit: any truly functional vat would need to be a surrogate body under the brain's control, and the resulting system would be an autonomous embodied agent rather than a disembodied brain.[cite:4][cite:7]

## Where the argument appears

The clearest formulation of the surrogate-body claim appears in "Embodiment or Envatment? Reflections on the Bodily Basis of Consciousness", a chapter by Cosmelli and Thompson in *Enaction: Toward a New Paradigm for Cognitive Science* (MIT Press, 2010).[cite:4][cite:7] The companion article, "Brain in a Vat or Body in a World? Brainbound versus Enactive Views of Experience", was published in *Philosophical Topics* 39(1) in 2011 and extends the same line of argument in a more formal philosophical register.[cite:1][cite:13]

Both texts are useful, but they do slightly different jobs. The 2010 chapter gives the most memorable statement of the null hypothesis. The 2011 paper adds a stronger discussion of supervenience, explanatory separability, and the limits of the orthodox neural-correlate framework.[cite:4][cite:10]

## The target of the critique

Thompson and Cosmelli are arguing against what they call the **brainbound** view. On that picture, the minimal constitutive basis of conscious experience is wholly neural, while body and environment contribute only causal inputs or enabling conditions. The brain-in-a-vat thought experiment is meant to support exactly this idea: if the brain received the right inputs, it could in principle enjoy the same experiences without any real body.[cite:10][cite:2]

Their approach is notable because they do not mainly attack the BIV as inconceivable or logically confused. Instead, they ask what would actually be required to build such a system in biologically realistic terms. That shift from intuition pump to implementation problem is what drives the argument.[cite:10][cite:1]

## **First requirement: keeping the brain alive**

A brain in a vat would first need an artificial system capable of sustaining its metabolism. That means more than bathing the brain in nutrients. It would require something functionally analogous to circulation, oxygen and glucose delivery, cerebrospinal regulation, ionic balance, and neuroendocrine coordination.[cite:7][cite:10]

The important point is that these processes are not static. The brain's requirements change continuously with its own intrinsic activity. A viable support system would therefore need to respond dynamically to the brain in real time. Thompson and Cosmelli argue that such a responsive, self-maintaining system already starts to look less like a passive vat and more like a body.[cite:4][cite:7]

## **Second requirement: preserving bodily self-regulation**

The argument then becomes stronger. The authors stress that the brain is not simply an input-output device waiting to be driven from outside. Much neural activity is endogenous and tied to the organism's ongoing self-maintenance. They draw on work associated with Antonio Damasio and Jaak Panksepp to argue that background consciousness depends on life-regulation processes involving brainstem and hypothalamic systems, affective circuitry, and extensive bodily feedback loops.[cite:7][cite:10]

On this view, what matters for creature consciousness is not only cortical content generation but also the organism's global homeodynamic organisation. If an envatted brain is to preserve anything like embodied subjectivity, the vat must recreate those bodily loops rather than merely stimulate a few sensory channels. The realising system for consciousness becomes brain-plus-surrogate-body, not brain alone.[cite:7][cite:4]

## **Third requirement: sensorimotor coupling**

Even if the brain were metabolically sustained and homeodynamically stabilised, that would still not generate ordinary world-directed experience. A further system would be needed to deliver structured sensory stimulation while remaining coupled to the brain's own activity, including motor commands, exploratory movements, and reafferent feedback.[cite:10]

This is where the enactive point becomes especially sharp. Experience is not passively received; it depends on active sensorimotor engagement. So a successful vat must not merely feed in pre-scripted signals. It must participate in reciprocal loops with the brain, updating sensory consequences as the brain acts. At that stage, the vat is functioning as an embodied agent in a world, whether physical or artificial.[cite:10][cite:9]

## **The null hypothesis**

The 2010 chapter states the conclusion plainly: any truly functional vat would need to be a surrogate body subject to control by the brain, understood as a self-regulating system with internal homeodynamic processes and the capacity for sensorimotor coupling with the outside world. In that sense, the so-called vat would be no vat at all, but some kind of autonomous embodied agent.[cite:4][cite:7]

This is important because their use of **body** is functional, not narrowly biological. They are not claiming that only ordinary organic tissue could realise consciousness. They are claiming that whatever realises it must instantiate the organisational features of a body: self-regulation, autonomy, and ongoing sensorimotor coupling.[cite:2][cite:10]

## Why this matters philosophically

The 2011 article goes beyond biological detail and argues that the BIV also fails as a supervenience thought experiment. The orthodox inference is supposed to work by holding the neural basis fixed while varying bodily and environmental conditions, thereby showing that experience depends only on the brain. Thompson and Cosmelli argue that this depends on an assumption of separability that the biology does not support.[cite:10]

Neural and extra-neural variables appear to be so densely coupled that they cannot simply be unplugged from one another while everything relevant remains unchanged. If that is right, the thought experiment does not successfully isolate a neural core realiser. It therefore cannot bear the explanatory weight that the brainbound view places on it.[cite:10][cite:2]

## Place in enactivism

This argument sits squarely within the wider enactive research programme associated with Varela, Thompson, and Rosch. The broader claim of enactivism is that cognition and experience are enacted through embodied engagement rather than produced entirely inside the head as internal representations of an independent world.[cite:6][cite:9][cite:15]

The BIV papers are distinctive because they turn one of philosophy of mind's favourite internalist thought experiments against itself. Instead of rejecting the scenario outright, Thompson and Cosmelli push it until it reveals its own hidden dependence on embodiment.[cite:1][cite:7]

## Relevance to uploading and emulation

Although these papers are not directly about whole-brain emulation, their argument has obvious implications for it. If creature consciousness depends on a dynamically coupled organismic system rather than on neural structure alone, then an upload that preserves only neural organisation would be incomplete unless it also recreated the relevant bodily and environmental loops.[cite:16][cite:18]

That does not rule out non-biological realisation. It does, however, suggest that successful emulation would need a surrogate body or body-like architecture rather than a free-floating computational brain. In that respect, the Thompson-Cosmelli argument is one of the most explicit enactive precursors to later embodiment critiques of mind uploading.[cite:16][cite:18]

## A concise way to read the paper

A good way to read Thompson and Cosmelli is to track a single progression:

- A brain needs more than nutrients to remain viable.[cite:7] [cite:10]
- Viability depends on dynamic self-regulation tied to bodily processes.[cite:4] [cite:7]
- Consciousness depends not only on neural contents but also on organism-level homeodynamic organisation.[cite:7] [cite:10]
- World-directed experience requires active sensorimotor coupling, not passive signal injection.[cite:9] [cite:10]
- Therefore, a functional vat must take on the role of a body.[cite:4] [cite:7]

## Key works

| Work                                                                                     | Role in the argument                                                                                |
|------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| <i>Embodiment or Envatment? Reflections on the Bodily Basis of Consciousness</i>         | Best source for the explicit surrogate-body formulation. [cite:4] [cite:7]                          |
| <i>Brain in a Vat or Body in a World? Brainbound versus Enactive Views of Experience</i> | Best source for the anti-brainbound and anti-supervenience framing.[cite:1] [cite:10]               |
| <i>Mind in Life</i>                                                                      | Broader philosophical and biological background for Thompson's enactive position.[cite:3] [cite:36] |
| <i>The Embodied Mind</i>                                                                 | Earlier foundational statement of the enactive approach. [cite:6] [cite:28]                         |

## Reading note

If the main goal is to harvest a quote and its argumentative setting, start with the 2010 chapter for the memorable formulation, then move to the 2011 *Philosophical Topics* paper for the deeper argument about why the thought experiment fails once neural and bodily dynamics are treated as nonseparable.[cite:4] [cite:10]